

Bacterial Surface Markers in Blood — Evidence Dossier (Updated)

This dossier summarizes bacterial-specific, surface-exposed markers detectable in blood with diagnostic and therapeutic potential in sepsis and invasive infections. It integrates classical markers (LPS, LTA, PGN) with additional targets (autolysins, OMPs, porins, WTA, sortase-anchored proteins, capsules, pili/flagella, biofilm components).

Top validated blood-detectable markers (clinical/translational evidence):

Marker	Why bacterial-specific	Clinical/Diagnostic use	Therapeutic potential
LPS (Lipid A / O-antigen)	Gram-negative outer membrane; absent in eukaryotes	Endotoxin assays (LAL), MS, species serotyping	Endotoxin neutralizers, LPS vaccines
Peptidoglycan fragments (MDP)	Universal bacterial cell wall component	ELISA/LC-MS; NOD2 biomarker	PGN sensing modulators, decoys
Lipoteichoic acid (LTA)	Gram-positive cell-wall glycolipid	ELISA; plasma marker for Gram+ bacteremia	Anti-LTA mAbs, imaging agents
Capsular polysaccharides (CPS)	Species/serotype-specific capsule	Rapid antigen tests (urine/serum), ELISA	Conjugate vaccines, capsule inhibitors
Flagellin (FliC)	Structural protein of bacterial flagella	ELISA for motile bacteria	Anti-flagellin vaccines
Outer membrane proteins & Porins	Proteins unique to Gram-negative OM	OM proteomics; antibody assays	Vaccine antigens; OMP-targeting peptides
Autolysins (murein hydrolases)	Peptidoglycan-remodeling enzymes	Enzyme assays; antibody detection	Autolysin inhibitors (antivirulence)
Wall teichoic acids (WTA)	Anionic glycopolymers in Gram-positive	Species-specific antigen detection; MS	WTA biosynthesis inhibitors
Sortase-anchored proteins (e.g., Poreins)	Surface-anchored Gram-positive surface proteins	Immunopassays, flow cytometry	Sortase inhibitors; antibody therapies
Pili/Fimbriae & Adhesins	Bacterial-specific appendages	Glycan arrays; antigen detection	Anti-adhesion therapeutics (mannosides)
Biofilm matrix components (Alginates, exopolysaccharides)	Unique extracellular polymers in bacteria	Optical sensors; matrix antigen detection	Matrix-disrupting enzymes; anti-biofilm drugs
S-layer proteins	Crystalline surface arrays in many bacteria	ELISA/immunoassays for species ID	Vaccine platforms; assembly inhibitors

Recommended diagnostic panel (near-term, evidence-backed):

1) LPS (endotoxin) assay — Gram-negative detection (LAL/MS confirmation). 2) Peptidoglycan fragment assay (MDP/PGN LC-MS or ELISA) — pan-bacterial detection. 3) Capsular antigen tests — pathogen-specific rapid ID (e.g., pneumococcal BinaxNOW). 4) LTA ELISA — Gram-positive support marker (*S. aureus*, *S. pneumoniae*). 5) Optional: Flagellin or OMV proteomics for added sensitivity to motile pathogens or species-specific protein signatures.

Pre-analytical & analytical considerations:

- Blood matrix interference: LPS and other markers may be masked by serum proteins; sample prep and inhibitor removal are crucial.
- Timing: PGN fragments and soluble proteins can be transient — early sampling improves sensitivity.
- Multiplexing: Combining broad (PGN) and specific (CPS, LPS) markers improves diagnostic accuracy.
- Validation: Use culture-confirmed sepsis cohorts to validate sensitivity/specificity and clinical cutoffs.

Suggested next steps for translation:

- Assemble retrospective plasma/serum samples from culture-proven Gram-positive and Gram-negative sepsis cases.
- Run LPS (LAL/MS), PGN LC-MS/ELISA, and CPS rapid tests in parallel to generate ROC curves.
- Validate in prospective cohorts; refine pre-analytics; pursue regulatory pathway for diagnostic approval.

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Note: Specific literature citations and assay protocols can be appended on request.